K-Means vs GMM: Hard vs Soft Clustering

Machine Learning and Neural Networks — Individual Tutorial
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GitHub Repository: https://github.com/mohammedabduljilani07/ML-kmeans-vs-gmm.git

Introduction

One of the most significant learning tasks (unsupervised) is clustering. We seek natural groups of data illustrated by no label data. K-Means and Gaussian Mixture Models (GMM) are two popular algorithms. Both partition data into K groups, but they are essentially different: K-Means makes use of hard assignment, and GMM makes use of soft assignment. As explained in this tutorial, the concept of difference is illustrated with the example of the Iris data set and clarifies when each of the methods is more suitable.

What is Clustering?

Clustering groups is the process of classifying points of data whereby grouped data points are more similar to each other than they are to those of other groups. There are no labels like there are in classification and the algorithm has to find structure itself. Among many other applications, clustering has been applied in customer segmentation, anomaly detection and analysis of biological data.

K-Means: Hard Assignment

K-Means is used to reduce the Within-Cluster Sum of Squares (WCSS) which refers to the sum of squared distances between each point and their assigned centroid. The algorithm switches between the assignment of every point to its closest centroid and the relocation of every centroid to the average of its assigned points. This will be done until assignments cease to be changed. Hard assignment is the critical property of K-Means. Each point is labeled with an individual value. The point located on the edge of two clusters is pushed to one cluster with total confidence which can be illusory. K-Means makes the assumption that clusters are approximately spherical and equal owing to its usage of Euclidean distance. K-Means can generate wrong boundaries in case of different clusters in shape or size. The elbow method can be used to choose K, by plotting WCSS versus K and indicates where the improvement starts leveling off, and the silhouette score, which is the level of fitment of each point to its respective cluster, compared with its neighbouring clusters. The score of the silhouettes above 1 refers to clearer clusters.

Gaussian Mixture Models: Soft Assignment

GMM uses a probabilistic strategy, which holds that the data was produced as a set of mixtures of K Gaussian distributions, with all having their means and covariance. Training is performed based on Expectation-Maximisation (EM) algorithm. The E-step refers to the computation of the probability of

every point to each component. The M-step is used to update the parameters of the individual Gaussian to maximise the likelihood. This is continued until convergence. GMM generates soft assignment as compared to K-Means. Probabilities are assigned to every point in respect to each cluster and a point towards a boundary can be a member of several clusters. As an example, one point may have a probability of 0.6 in one cluster and 0.4 in the other. This is a true depiction of groping in the dark that K-Means does not take into account. The Bayesian Information Criterion (BIC) or the Akaike Information Criterion (AIC) is used to choose K where the good fit is rewarded and complexity unnecessarily penalised. Small values represent an improved model.

Demonstration: Iris Dataset

The number of variables of the Iris data set is four and the sample of the flower is 150, with the variables being sepal length, sepal width, petal length, and petal width. The three species, namely Setosa, Versicolor, and Virginica are the samples. The algorithms do not use the true labels and are evaluated using them only.

Figure 1: Iris Dataset — True Species Labels

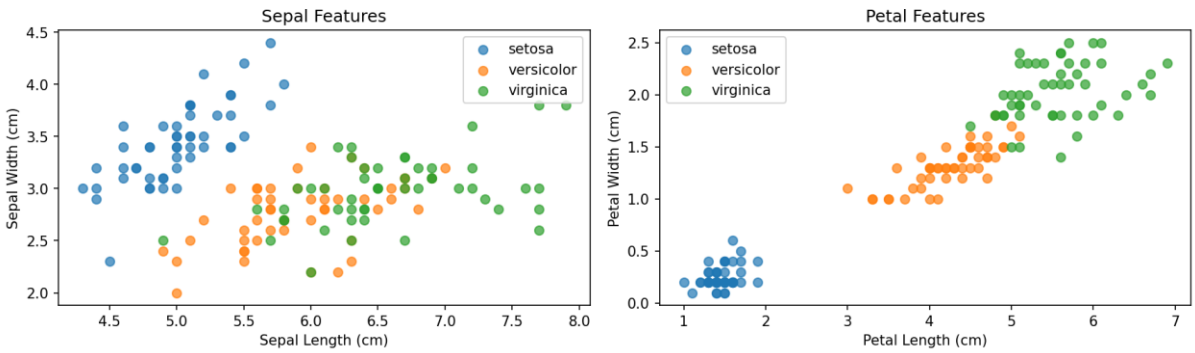


Figure 1 shows the raw data. Setosa is distinctly defined as it is different to the other two species. Versicolor and Virginica have a minor overlap particularly in sepal feature space. It is in this overlapping that soft assignment comes into significance. StandardScaler was used to standardise all features to avoid the dominance of one feature over the rest because of scale.

Figure 2: True Labels vs K-Means vs GMM (PCA reduced)

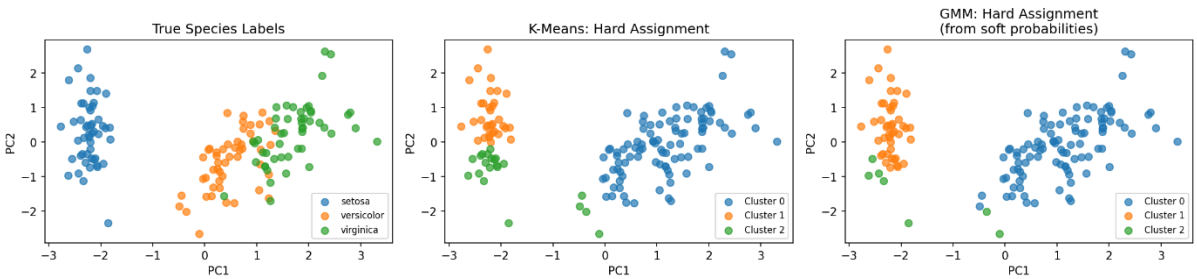
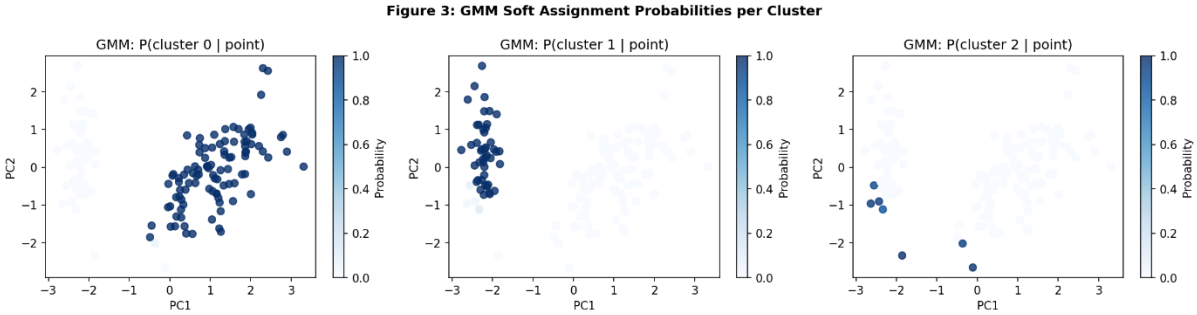
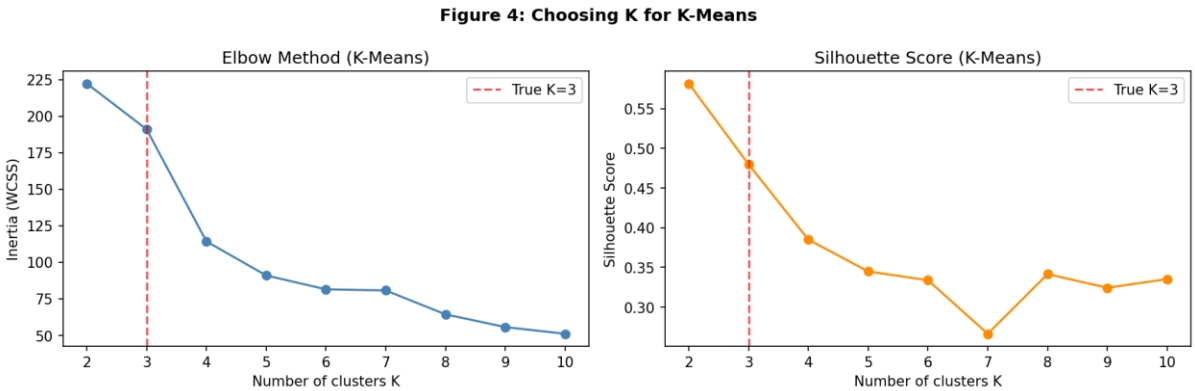
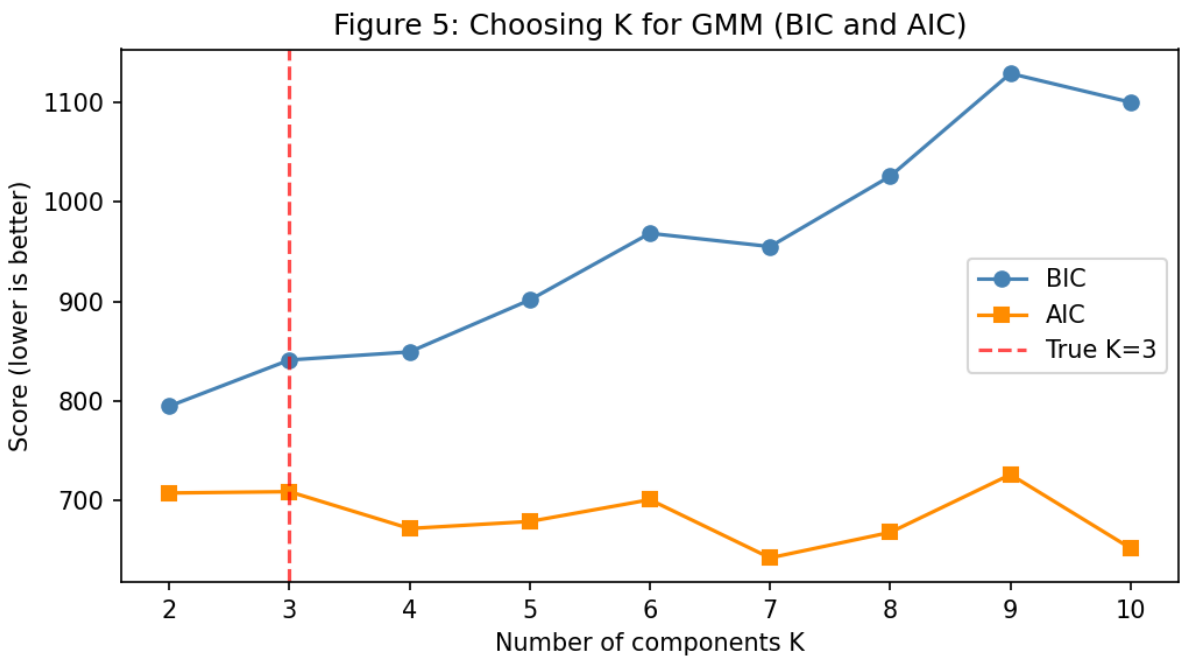


Figure 2 displays the results of clustering; the results were obtained after they were reduced into two dimensions using PCA. Both K-Means and GMM are successful in separating Setosa, but variation is evident at the boundary of Versicolor and Virginica.



The soft probabilities of GMM are presented in figure 3. The region of overlap contains mixed probabilities e.g. a cluster receives 0.6 and the other one receives 0.4. K-Means cannot come up with such probabilities; all the points will be 0 or 1.





The K selection is illustrated in figure 4 and 5. K-Means In the K-Means case, the elbow-method and silhouette score suggest that K =3. In the case of GMM, BIC and AIC also attain their minimum at K=3 which assures the actual number of species.

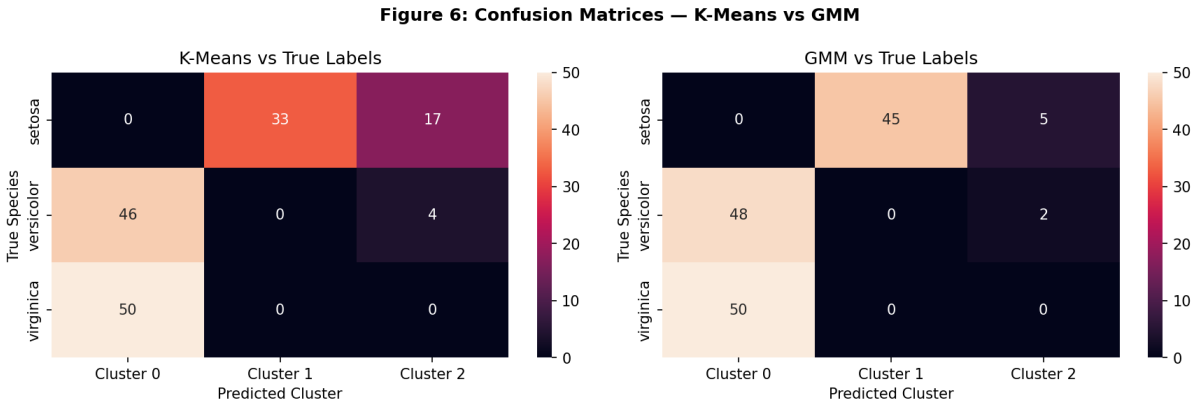


Figure 6 presents confusion matrices between both algorithms and the actual species. They both do equally well in difficult tasks with the majority of misclassifications found at the Versicolor/Virginica interface.

Comparison

K-Means is interpretable, simple and fast. It is efficient with clusters that are separated well and roughly spherical, and is hard assigned well when cluster labels are needed. GMM is more likely to be expressive and principled. It is suitable when clusters have overlaps, confidence scores are required or when the shapes of the clusters are dissimilar. Its soft assignment sincerely symbolizes indeterminacy at edges. Both of these require K to be given in advance, either can converge to local optima, and both of them are not very well-suited to non-linear cluster structures.

Conclusion

K-Means and GMM are the complements. The decision relies on the statistics and the need in question. K-Means can be used when clusters are distinctly separated and when speed and simplicity are required. GMM is preferable when estimates of uncertainty are important or there are clusters which overlap. Both algorithms on the Iris dataset identify comparable hard boundaries, yet only GMM identifies true points of uncertainty, that is useful in the real world.

References

Bishop, C.M. (2006). *Pattern Recognition and Machine Learning.* Springer.

MacQueen, J. (1967). Some methods for classification and analysis of multivariate observations. Proceedings of the 5th Berkeley Symposium, 1, 281–297.

Dempster, A.P., Laird, N.M., Rubin, D.B. (1977). Maximum likelihood from incomplete data via the EM algorithm. Journal of the Royal Statistical Society B, 39(1), 1–38.

scikit-learn. KMeans. https://scikit-learn.org/stable/modules/generated/sklearn.cluster.KMeans.html

scikit-learn. GaussianMixture. https://scikit-learn.org/stable/modules/generated/sklearn.mixture.GaussianMixture.html